MODULE 3 HEART

The Heart.

BLOCK 3 · MODULE 2: STRUCTURES, VALVES, AND THE PATHWAY OF BLOOD

A reference for the heart video and lab. This chapter covers the heart wall and pericardium, the four chambers and four valves, the fibrous skeleton, cardiac muscle tissue, the pathway of blood, and the great vessels and coronary circulation. The focus is on structure.

BY THE END

1. Name the layers of the heart wall and the pericardium.
2. Identify the four chambers and the four valves, and what each one separates.
3. Identify the internal features of each chamber.
4. Describe cardiac muscle tissue and the intercalated disc.
5. Trace the pathway of blood through the heart in order.
6. Name the great vessels and the coronary arteries and veins.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The heart

drsrennie-stack.github.io/new-build-bio4-solano/heart-concept-videos.html

The Heart, an Overview

The heart is a hollow, muscular pump about the size of a closed fist. It sits in the mediastinum and drives blood through two circuits, one to the lungs and one to the body.

LOCATION AND ORIENTATION

Structure	What it is
Heart	A hollow, muscular organ that pumps blood through the blood vessels
Location	In the mediastinum, between the two lungs, resting on the diaphragm and tilted to the left
Mass	About two thirds of the heart's mass lies to the left of the midline
Apex	The blunt, rounded inferior tip, pointing down, forward, and to the left. It is the tip of the left ventricle
Base	The broad superior surface, mostly left atrium, where the great vessels enter and leave
Anterior surface	The sternocostal surface, mostly right ventricle, lying under the sternum and ribs
Inferior surface	The diaphragmatic surface, mostly left ventricle, resting on the diaphragm
Pulmonary circuit	The loop the right side of the heart drives, carrying blood to the lungs and back
Systemic circuit	The loop the left side of the heart drives, carrying blood to the body and back

The Heart Wall and Pericardium

The heart is wrapped in a double sac, the pericardium, and its own wall is built in three layers.

THE PERICARDIUM

Structure	What it is
Pericardium	The double-walled sac that encloses the heart
Fibrous pericardium	The tough, inelastic outer layer of dense connective tissue. It anchors the heart in the mediastinum and prevents overfilling
Serous pericardium	The inner double membrane: a parietal layer lining the fibrous pericardium and a visceral layer on the heart surface
Pericardial cavity	The space between the two serous layers, holding a film of serous fluid that reduces friction as the heart beats

The three layers of the heart wall

COMPARE THEM FROM OUTSIDE IN

Layer	Position	Description
Epicardium	The outer layer	The visceral layer of the serous pericardium. A thin serous membrane on the heart surface, with fat and the coronary vessels running in it
Myocardium	The thick middle layer	Cardiac muscle tissue in spiralling bundles. About ninety five percent of the wall thickness, and the layer that contracts
Endocardium	The inner layer	A smooth endothelium lining the chambers and covering the valves, continuous with the lining of the great vessels

HOLD ONTO THIS

The epicardium and the visceral serous pericardium are the same membrane under two names. Which name you use depends on whether you are describing the heart wall or the sac.

The Chambers

The heart has four chambers: two thin-walled atria that receive blood and two thick-walled ventricles that pump it out.

WHAT EACH CHAMBER RECEIVES AND WHERE IT SENDS BLOOD

Chamber	Receives blood from	Pumps blood to
Right atrium	The body, through the superior and inferior venae cavae, and the heart wall through the coronary sinus	The right ventricle
Right ventricle	The right atrium	The lungs, through the pulmonary trunk
Left atrium	The lungs, through the four pulmonary veins	The left ventricle
Left ventricle	The left atrium	The body, through the aorta. Its wall is the thickest of the four chambers

Internal features

THE NAMED STRUCTURES INSIDE THE CHAMBERS

Structure	What it is
Interatrial septum	The wall between the two atria
Fossa ovalis	The oval depression in the interatrial septum, the remnant of the fetal foramen ovale
Interventricular septum	The thick wall between the two ventricles
Auricle	A small wrinkled pouch on each atrium that slightly increases its capacity
Pectinate muscles	Parallel muscular ridges on the walls of the atria and the auricles
Crista terminalis	The ridge in the right atrium separating the smooth posterior wall from the rough pectinate anterior wall
Sinus venarum	The smooth-walled posterior part of the right atrium, where the venae cavae empty
Trabeculae carneae	The irregular muscular ridges on the inner walls of the ventricles
Papillary muscles	Cone-shaped muscular projections of the ventricle wall that anchor the chordae tendineae
Moderator band	A muscular band in the right ventricle, also called the septomarginal trabecula, that carries part of the conduction system to the anterior papillary muscle
Conus arteriosus	The smooth outflow tract of the right ventricle, leading to the pulmonary valve
Aortic vestibule	The smooth outflow tract of the left ventricle, leading to the aortic valve

The Heart Valves

Four valves keep blood moving in one direction. Each opens to let blood pass and closes to stop it flowing back.

COMPARE THEM BY TYPE AND LOCATION

Valve	Type	Location	Prevents backflow into
Tricuspid valve	Atrioventricular, three cusps	Between the right atrium and the right ventricle	The right atrium
Mitral valve	Atrioventricular, two cusps; also called the bicuspid valve	Between the left atrium and the left ventricle	The left atrium
Pulmonary valve	Semilunar, three cusps	Between the right ventricle and the pulmonary trunk	The right ventricle
Aortic valve	Semilunar, three cusps	Between the left ventricle and the aorta	The left ventricle

WHAT HOLDS THE ATRIOVENTRICULAR VALVES SHUT

Structure	What it is
Chordae tendineae	Strong fibrous cords that tie the cusps of an atrioventricular valve to the ventricle wall
Papillary muscles	The muscular projections that anchor the chordae tendineae and hold the valves shut against the pressure of a contracting ventricle
Fibrous skeleton	The rings of dense connective tissue around the four valve openings. It anchors the valves, gives the cardiac muscle a point of attachment, and electrically insulates the atria from the ventricles

HOLD ONTO THIS

Only the atrioventricular valves have chordae tendineae and papillary muscles. The semilunar valves have neither, because their pocket-shaped cusps fill from above and close themselves. If a question mentions chordae, the answer is tricuspid or mitral.

Cardiac Muscle Tissue

The myocardium is built of cardiac muscle, a tissue found nowhere else in the body. Its structure lets the whole heart contract as one coordinated unit.

CARDIAC MUSCLE AT THE CELLULAR LEVEL

Structure	What it is
Cardiac muscle	The involuntary, striated muscle tissue that forms the myocardium
Cardiomyocyte	A cardiac muscle cell, short and branching, usually with a single central nucleus
Striations	Cardiac muscle is striped, like skeletal muscle, because it too is built of ordered sarcomeres
Intercalated discs	The specialised junctions where neighbouring cardiomyocytes meet end to end
Desmosomes	Anchoring junctions within the intercalated discs that keep the cells from pulling apart
Gap junctions	Channels within the intercalated discs that let the contraction signal pass directly from cell to cell
Abundant mitochondria	Cardiac muscle is packed with mitochondria, about a quarter of the cell volume, to fuel its constant lifelong work

The Pathway of Blood

Follow one drop of blood through the heart, in order. It makes a full loop: into the right side, out to the lungs, back into the left side, and out to the body.

1. **Right atrium** oxygen-poor blood returns from the body through the superior and inferior venae cavae
2. **Tricuspid valve** blood passes from the right atrium down into the right ventricle
3. **Right ventricle** contracts and pumps the blood upward and out
4. **Pulmonary valve** blood passes into the pulmonary trunk, which branches to the lungs

5. **Lungs** blood picks up oxygen and releases carbon dioxide, then returns through the pulmonary veins
6. **Left atrium** receives the now oxygen-rich blood from the pulmonary veins
7. **Mitral valve** blood passes from the left atrium down into the left ventricle
8. **Left ventricle** the thickest chamber contracts and pumps the blood out with force
9. **Aortic valve** blood passes into the aorta and out to the body, completing the loop

The Great Vessels and Coronary Circulation

The great vessels are the large pipes attached to the base of the heart. The heart wall has its own blood supply, the coronary circulation.

THE GREAT VESSELS

Structure	What it is
Superior vena cava	Returns oxygen-poor blood from the head, neck, and upper limbs to the right atrium
Inferior vena cava	Returns oxygen-poor blood from the trunk and lower limbs to the right atrium
Pulmonary trunk	Carries blood from the right ventricle and branches into the right and left pulmonary arteries
Pulmonary veins	Four veins that return oxygen-rich blood from the lungs to the left atrium
Aorta	The largest artery, carrying oxygen-rich blood from the left ventricle to the body

The coronary arteries

The two coronary arteries leave the base of the aorta, run in the grooves on the heart's surface, and branch over the myocardium. The flow is aorta, coronary arteries, capillaries of the myocardium, cardiac veins, coronary sinus, right atrium.

CORONARY ARTERIES AND WHAT EACH SUPPLIES

Coronary artery	Main branches	Region supplied
Right coronary artery (RCA)	Right marginal artery; posterior interventricular artery, the posterior descending or PDA	Right atrium, most of the right ventricle, part of the left ventricle, the posterior septum, and usually the SA and AV nodes
Left coronary artery (LCA)	Anterior interventricular artery (LAD); circumflex artery; left marginal artery	Left atrium, left ventricle, anterior septum, and the apex

The coronary veins

VENOUS DRAINAGE OF THE HEART WALL

Structure	What it is
Coronary sinus	The large vein on the posterior heart that collects the cardiac veins and empties into the right atrium
Great cardiac vein	Runs with the anterior interventricular artery (LAD)
Middle cardiac vein	Runs with the posterior interventricular artery (PDA)
Small cardiac vein	Runs with the right marginal artery
Anterior cardiac veins	Drain the right ventricle directly into the right atrium, bypassing the coronary sinus

HOLD ONTO THIS

Every cardiac vein you can name runs alongside a coronary artery of the matching name. Learn the arteries and the veins come free, because they travel in the same grooves.

STRUCTURE EXPLAINS THE EMERGENCY

The **coronary arteries** are the heart's own supply line, and they are narrow. When one is blocked, the patch of myocardium beyond it is starved of oxygen and its muscle begins to die. That is a heart attack, a myocardial infarction, and it is why the coronary vessels matter far beyond their small size.

A **valve** that does not close cleanly lets blood slip backward, and the turbulence is heard through a stethoscope as a murmur. The **chordae tendineae** and **papillary muscles** are what hold the atrioventricular valves shut against the force of a contracting ventricle; if a papillary muscle is damaged, the valve can fail.

Fluid building up in the **pericardial cavity**, called cardiac tamponade, squeezes the heart from outside and limits how much it can fill. The same fluid film that normally lets the heart glide can, in excess, stop it from working.